FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

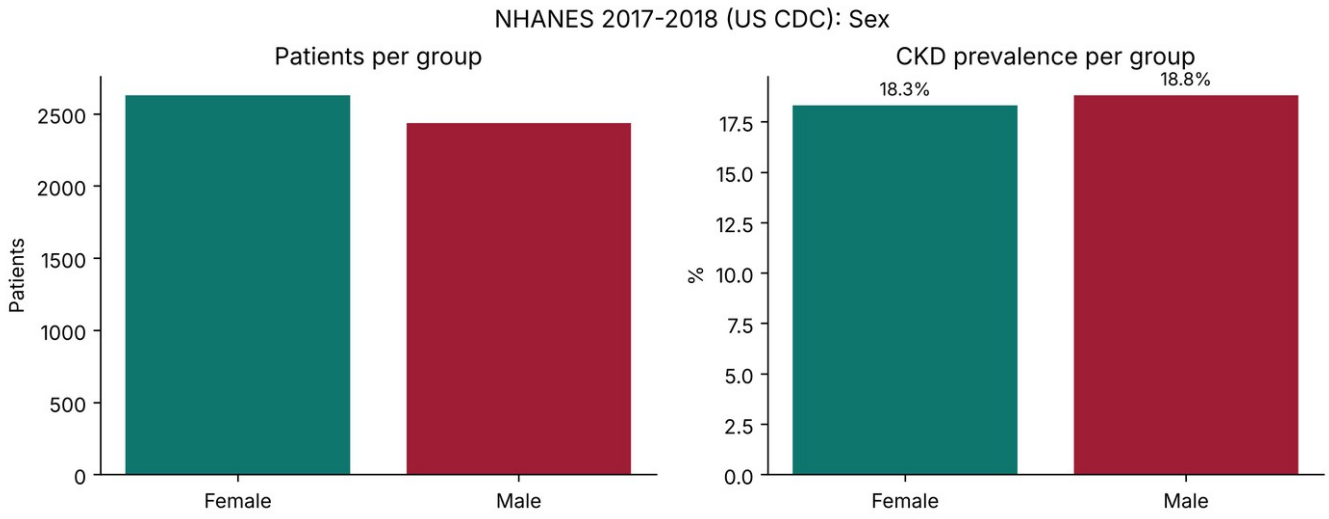
	Value
Dataset	NHANES 2017-2018 (US CDC)
Feature set	Routine blood tests (17 features, no kidney-function tests)
Sensitive attribute	Sex
Patients	5063
Features	17
Groups	Female, Male
Decision threshold	training-set CKD prevalence
CV protocol	stratified 5-fold, 5 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **Random Forest** (accuracy 0.737, balanced accuracy 0.737, AUC 0.806).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **Female** group (Equal Opportunity gap 0.055–0.088).
• Even a perfect classifier has DP diff = 0.005 and DI = 0.974 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.055 → 0.011, balanced accuracy 0.735 → 0.726.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **RW + TO (hybrid)**, Equalized Odds gap 0.069 → 0.012, balanced accuracy 0.735 → 0.727.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.044 → **0.031** compared with accuracy-driven model selection, balanced accuracy 0.734 → 0.727.
• Statistical significance: 2 of 10 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.740–0.749, Equalized Odds gap 0.021–0.033; best mitigated gap 0.016.

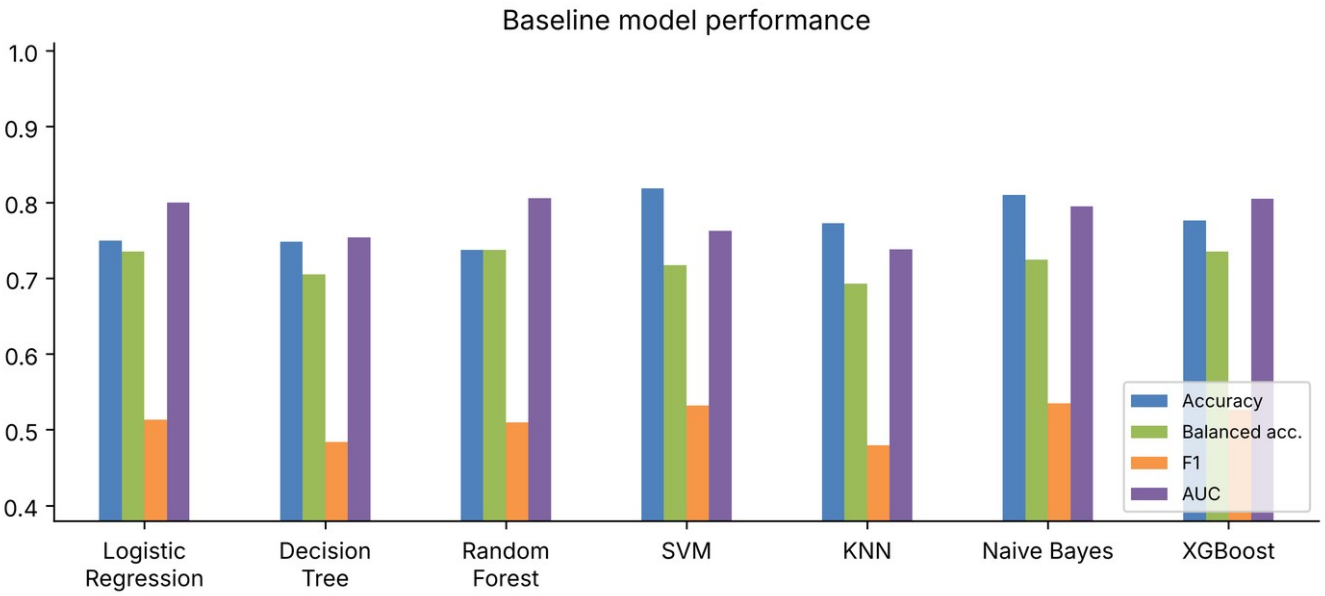
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Female	2627	51.9	481	18.3	0.170
Male	2436	48.1	458	18.8	0.130
All	5063	100	939	18.5	0.150


NHANES 2017-2018 (US CDC): Sex
Patients per group
Patients
2500
2000
1500
1000
500
0
Female
Male
CKD prevalence per group
%
18.3%
18.8%
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Female
Male

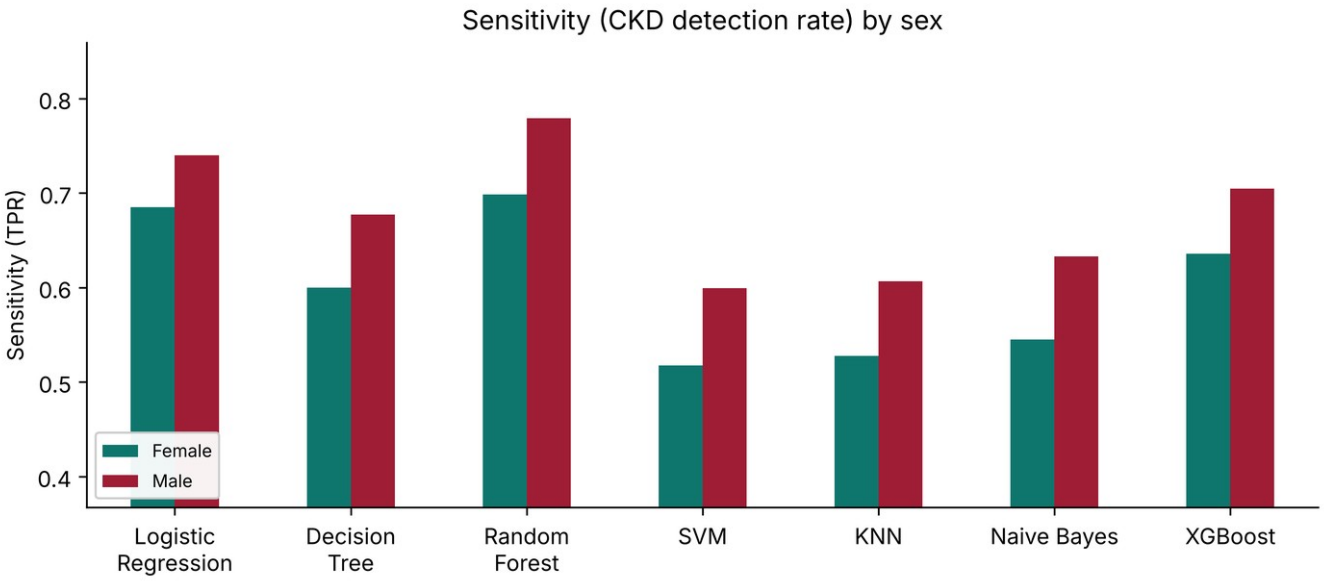
2. Baseline model performance

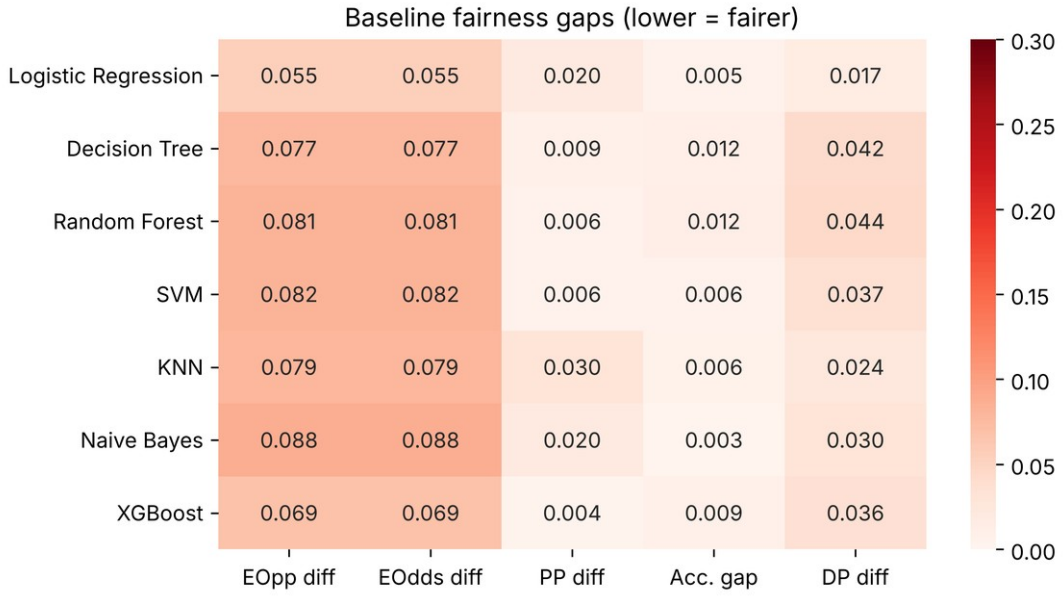
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.750 ± 0.002	0.402 ± 0.002	0.712 ± 0.002	0.513 ± 0.002	0.800 ± 0.001	0.735 ± 0.002
Decision Tree	0.748 ± 0.010	0.390 ± 0.010	0.637 ± 0.023	0.484 ± 0.004	0.754 ± 0.006	0.705 ± 0.004
Random Forest	0.737 ± 0.002	0.390 ± 0.002	0.738 ± 0.002	0.510 ± 0.002	0.806 ± 0.004	0.737 ± 0.002
SVM	0.818 ± 0.002	0.509 ± 0.005	0.557 ± 0.004	0.532 ± 0.002	0.763 ± 0.002	0.717 ± 0.001
KNN	0.772 ± 0.002	0.416 ± 0.004	0.566 ± 0.005	0.480 ± 0.003	0.738 ± 0.002	0.693 ± 0.002
Naive Bayes	0.810 ± 0.001	0.490 ± 0.003	0.588 ± 0.003	0.534 ± 0.003	0.795 ± 0.001	0.724 ± 0.002
XGBoost	0.776 ± 0.005	0.433 ± 0.008	0.669 ± 0.008	0.526 ± 0.009	0.805 ± 0.002	0.735 ± 0.006



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.017 ± 0.004	0.948 ± 0.011	0.055 ± 0.006	0.055 ± 0.006	0.020 ± 0.005	0.005 ± 0.004
Decision Tree	0.042 ± 0.010	0.872 ± 0.030	0.077 ± 0.017	0.077 ± 0.017	0.009 ± 0.003	0.012 ± 0.006
Random Forest	0.044 ± 0.003	0.882 ± 0.007	0.081 ± 0.014	0.081 ± 0.014	0.006 ± 0.005	0.012 ± 0.005
SVM	0.037 ± 0.003	0.835 ± 0.013	0.082 ± 0.007	0.082 ± 0.007	0.006 ± 0.004	0.006 ± 0.002
KNN	0.024 ± 0.003	0.910 ± 0.012	0.079 ± 0.012	0.079 ± 0.012	0.030 ± 0.007	0.006 ± 0.003
Naive Bayes	0.030 ± 0.003	0.874 ± 0.011	0.088 ± 0.007	0.088 ± 0.007	0.020 ± 0.006	0.003 ± 0.003
XGBoost	0.036 ± 0.004	0.881 ± 0.013	0.069 ± 0.009	0.069 ± 0.009	0.004 ± 0.003	0.009 ± 0.003
Perfect classifier (reference)	0.005	0.974	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by sex
Sensitivity (TPR)
0.8
0.7
0.6
0.5
0.4
Female
Male
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.055 0.055 0.020 0.005 0.017
Decision Tree 0.077 0.077 0.009 0.012 0.042
Random Forest 0.081 0.081 0.006 0.012 0.044
SVM 0.082 0.082 0.006 0.006 0.037
KNN 0.079 0.079 0.030 0.006 0.024
Naive Bayes 0.088 0.088 0.020 0.003 0.030
XGBoost 0.069 0.069 0.004 0.009 0.036
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.30
0.25
0.20
0.15
0.10
0.05
0.00

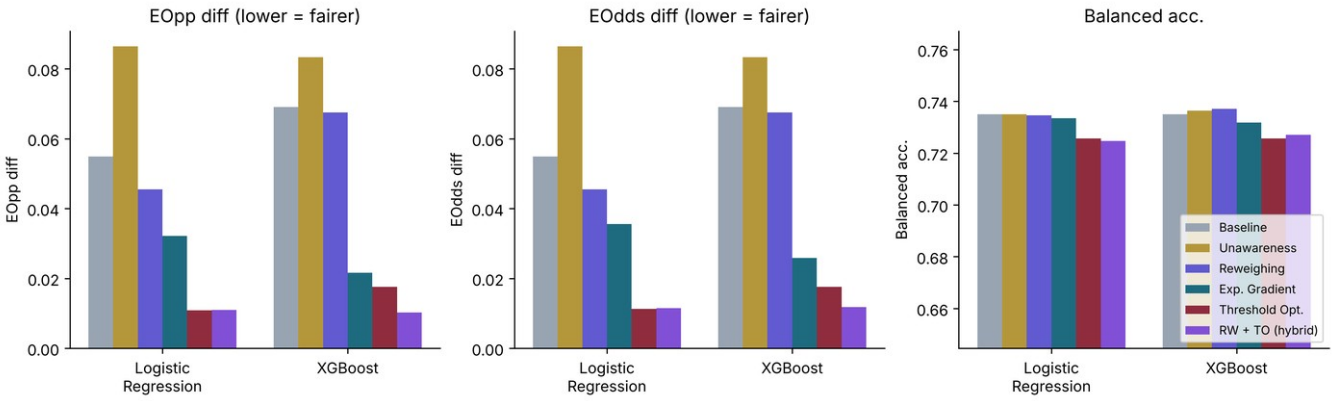
4. Bias mitigation

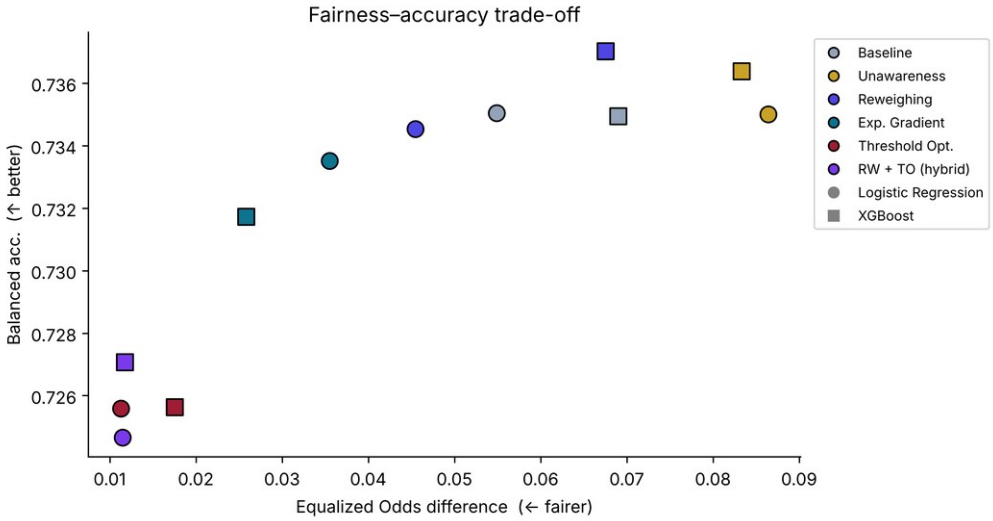
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.750 ± 0.002	0.735 ± 0.002	0.712 ± 0.002	0.513 ± 0.002
Logistic Regression	Unawareness	0.750 ± 0.001	0.735 ± 0.001	0.711 ± 0.002	0.514 ± 0.001
Logistic Regression	Reweighing	0.750 ± 0.002	0.735 ± 0.002	0.710 ± 0.002	0.513 ± 0.002
Logistic Regression	Exp. Gradient	0.750 ± 0.002	0.734 ± 0.003	0.708 ± 0.004	0.512 ± 0.003
Logistic Regression	Threshold Opt.	0.785 ± 0.004	0.726 ± 0.003	0.631 ± 0.009	0.521 ± 0.003
Logistic Regression	RW + TO (hybrid)	0.784 ± 0.004	0.725 ± 0.003	0.630 ± 0.009	0.520 ± 0.004
XGBoost	Baseline	0.776 ± 0.005	0.735 ± 0.006	0.669 ± 0.008	0.526 ± 0.009
XGBoost	Unawareness	0.776 ± 0.002	0.736 ± 0.002	0.674 ± 0.006	0.527 ± 0.002
XGBoost	Reweighing	0.777 ± 0.005	0.737 ± 0.006	0.674 ± 0.009	0.528 ± 0.008
XGBoost	Exp. Gradient	0.793 ± 0.003	0.732 ± 0.005	0.634 ± 0.008	0.532 ± 0.006
XGBoost	Threshold Opt.	0.776 ± 0.009	0.726 ± 0.002	0.646 ± 0.010	0.517 ± 0.007
XGBoost	RW + TO (hybrid)	0.779 ± 0.011	0.727 ± 0.003	0.645 ± 0.019	0.520 ± 0.007

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.055 ± 0.006	0.055 ± 0.006	0.020 ± 0.005	0.017 ± 0.004	0.948 ± 0.011	0.005 ± 0.004
Logistic Regression	Unawareness	0.086 ± 0.006	0.086 ± 0.006	0.006 ± 0.001	0.053 ± 0.003	0.850 ± 0.008	0.019 ± 0.001
Logistic Regression	Reweighing	0.045 ± 0.003	0.045 ± 0.003	0.025 ± 0.004	0.009 ± 0.003	0.972 ± 0.010	0.010 ± 0.003
Logistic Regression	Exp. Gradient	0.032 ± 0.013	0.036 ± 0.009	0.035 ± 0.005	0.007 ± 0.008	0.978 ± 0.024	0.019 ± 0.005
Logistic Regression	Threshold Opt.	0.011 ± 0.006	0.011 ± 0.005	0.012 ± 0.010	0.008 ± 0.005	0.970 ± 0.018	0.006 ± 0.002
Logistic Regression	RW + TO (hybrid)	0.011 ± 0.006	0.011 ± 0.005	0.011 ± 0.008	0.007 ± 0.005	0.974 ± 0.017	0.005 ± 0.002
XGBoost	Baseline	0.069 ± 0.009	0.069 ± 0.009	0.004 ± 0.003	0.036 ± 0.004	0.881 ± 0.013	0.009 ± 0.003
XGBoost	Unawareness	0.083 ± 0.008	0.083 ± 0.008	0.007 ± 0.005	0.047 ± 0.005	0.849 ± 0.016	0.015 ± 0.004
XGBoost	Reweighing	0.068 ± 0.011	0.068 ± 0.011	0.007 ± 0.005	0.036 ± 0.004	0.882 ± 0.012	0.009 ± 0.005
XGBoost	Exp. Gradient	0.022 ± 0.012	0.026 ± 0.004	0.025 ± 0.014	0.009 ± 0.007	0.968 ± 0.027	0.010 ± 0.004
XGBoost	Threshold Opt.	0.018 ± 0.015	0.018 ± 0.015	0.023 ± 0.015	0.005 ± 0.002	0.981 ± 0.008	0.009 ± 0.009
XGBoost	RW + TO (hybrid)	0.010 ± 0.009	0.012 ± 0.008	0.016 ± 0.014	0.006 ± 0.002	0.978 ± 0.007	0.008 ± 0.006





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	0.045	[0.026, 0.066]	0.004	-0.001	[-0.006, 0.005]	0.848
Logistic Regression	Reweighing	-0.006	[-0.013, 0.005]	0.222	-0.001	[-0.003, 0.001]	0.620
Logistic Regression	Exp. Gradient	-0.018	[-0.047, 0.029]	0.380	-0.004	[-0.009, 0.001]	0.164
Logistic Regression	Threshold Opt.	-0.023	[-0.065, 0.033]	0.298	-0.009	[-0.019, -0.000]	0.046
Logistic Regression	RW + TO (hybrid)	-0.023	[-0.065, 0.032]	0.276	-0.009	[-0.019, -0.001]	0.036
XGBoost	Unawareness	0.012	[-0.014, 0.040]	0.372	0.001	[-0.006, 0.008]	0.730
XGBoost	Reweighing	0.001	[-0.026, 0.030]	0.970	0.002	[-0.005, 0.009]	0.560
XGBoost	Exp. Gradient	-0.048	[-0.083, -0.007]	0.032	-0.005	[-0.015, 0.005]	0.312
XGBoost	Threshold Opt.	-0.047	[-0.079, -0.004]	0.040	-0.009	[-0.017, -0.001]	0.032
XGBoost	RW + TO (hybrid)	-0.056	[-0.100, 0.015]	0.108	-0.008	[-0.018, 0.001]	0.094

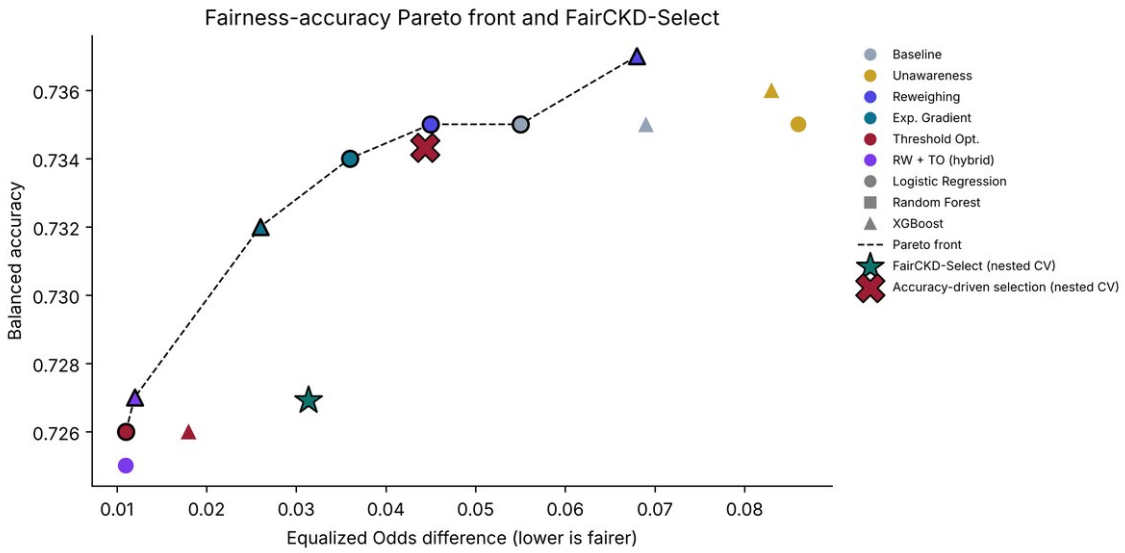
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.779 ± 0.017	0.727 ± 0.004	0.644 ± 0.018	0.031 ± 0.021	0.031 ± 0.021	0.012 ± 0.015
Accuracy-driven selection	0.763 ± 0.001	0.734 ± 0.004	0.688 ± 0.012	0.044 ± 0.002	0.044 ± 0.002	0.028 ± 0.002

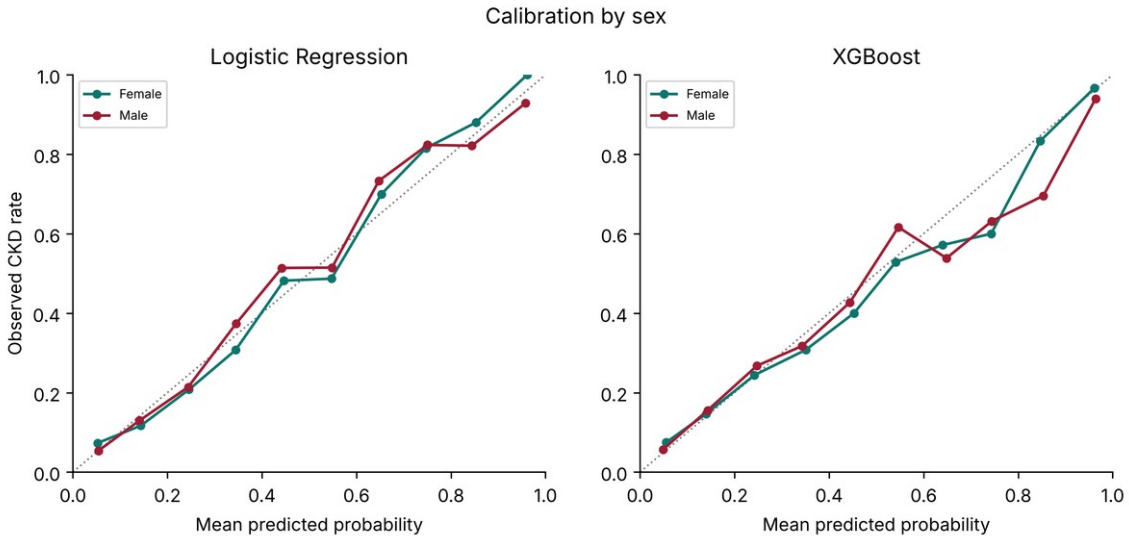
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
Logistic Regression	Exp. Gradient	1
Logistic Regression	RW + TO (hybrid)	1
Logistic Regression	Threshold Opt.	3
XGBoost	RW + TO (hybrid)	2
XGBoost	Threshold Opt.	3



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Female	2627	18.31	18.293	-0.017	0.028	0.112	Logistic Regression
Male	2436	18.801	18.795	-0.007	0.016	0.111	Logistic Regression
Female	2627	18.31	17.817	-0.493	0.019	0.112	XGBoost
Male	2436	18.801	18.687	-0.114	0.020	0.113	XGBoost

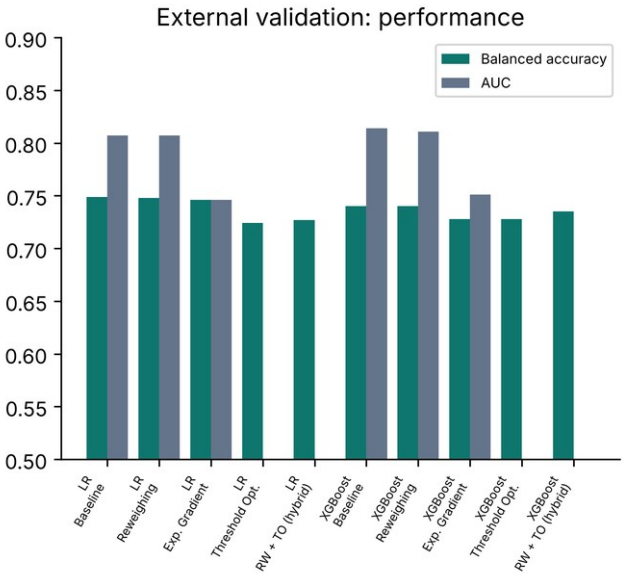


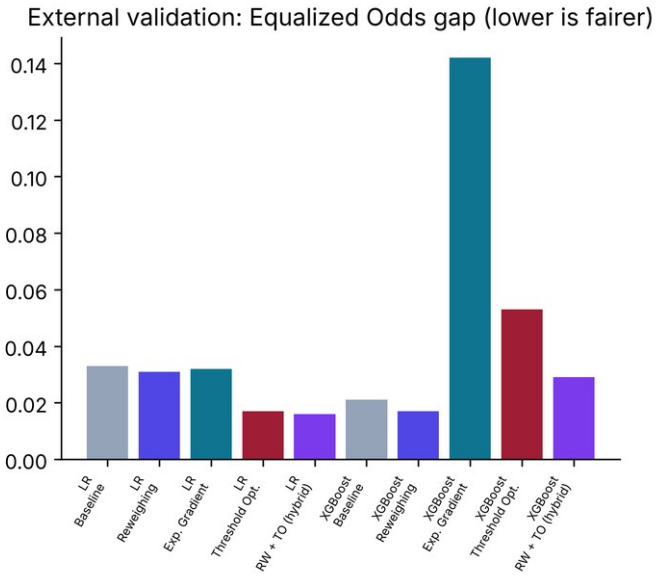
4d. Population-weighted estimates (NHANES survey weights)

Model	W Accuracy	W Balanced acc.	W EOpp diff	W EOdds diff	W DP diff	W TPR [Female]	W TPR [Male]
Logistic Regression	0.787	0.731	0.029	0.029	0.008	0.641	0.670
Decision Tree	0.786	0.699	0.035	0.035	0.012	0.570	0.591
Random Forest	0.781	0.741	0.080	0.080	0.015	0.649	0.730
SVM	0.850	0.702	0.064	0.064	0.017	0.468	0.532
KNN	0.805	0.690	0.093	0.093	0.006	0.490	0.583
Naive Bayes	0.847	0.713	0.058	0.058	0.014	0.502	0.560
XGBoost	0.810	0.723	0.063	0.063	0.025	0.575	0.638

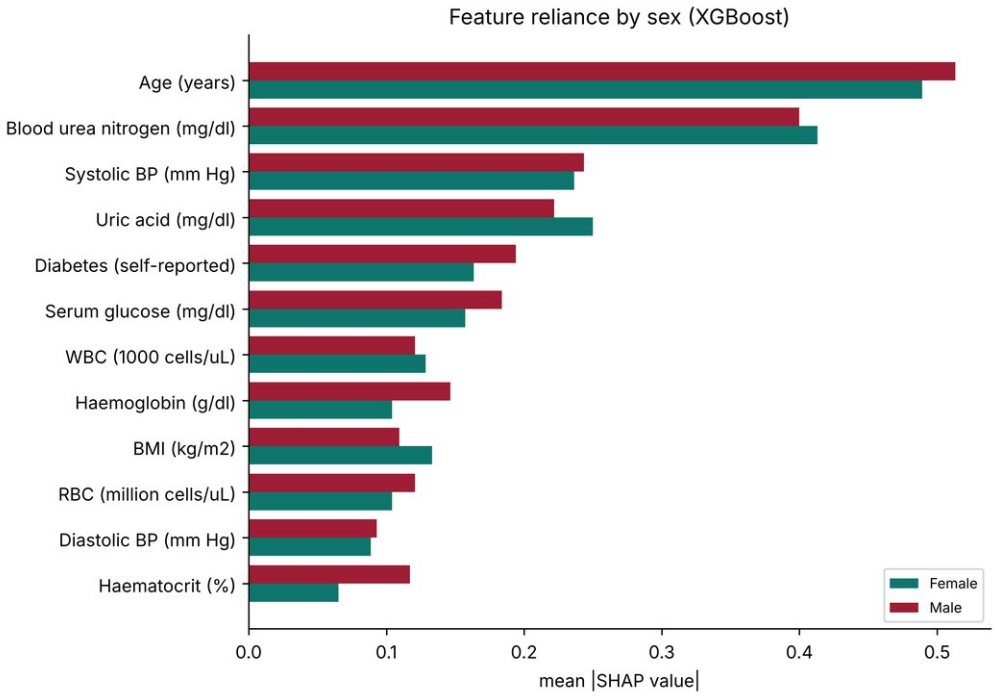
4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

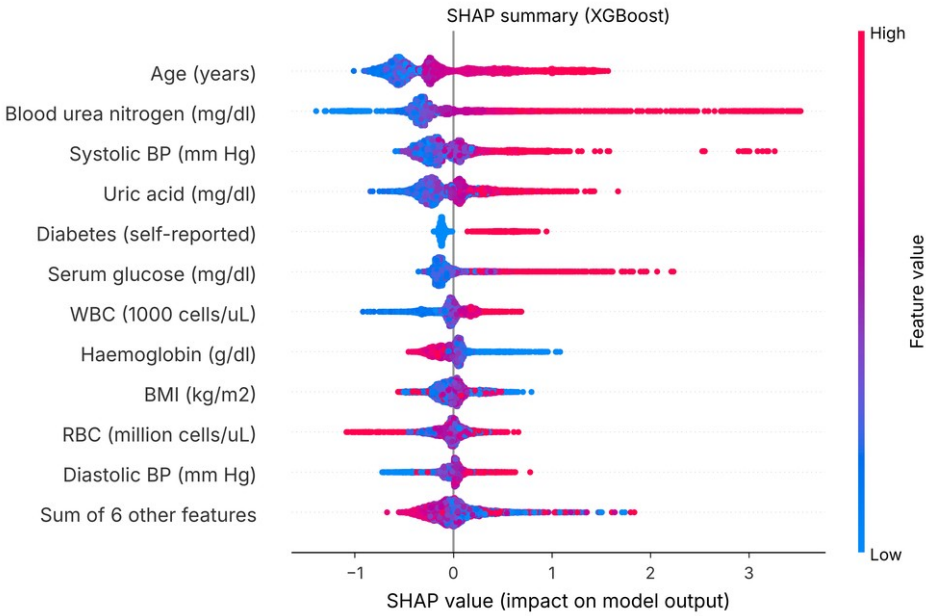
Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.784	0.749	0.696	0.807	0.033	0.033
Logistic Regression	Reweighing	0.783	0.748	0.696	0.807	0.019	0.031
Logistic Regression	Exp. Gradient	0.773	0.746	0.705	0.746	0.013	0.032
Logistic Regression	Threshold Opt.	0.827	0.724	0.571		0.003	0.017
Logistic Regression	RW + TO (hybrid)	0.827	0.727	0.578		0.002	0.016
XGBoost	Baseline	0.791	0.740	0.666	0.814	0.007	0.021
XGBoost	Reweighing	0.790	0.740	0.666	0.811	0.012	0.017
XGBoost	Exp. Gradient	0.804	0.728	0.615	0.751	0.142	0.142
XGBoost	Threshold Opt.	0.800	0.728	0.622		0.053	0.053
XGBoost	RW + TO (hybrid)	0.786	0.735	0.658		0.029	0.029





6. Explainability (SHAP)





Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.